# Protik Dev

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## Summary

Software Engineer with 5+ years of experience building AI features, third-party API integrations, and backend services in TypeScript and Node.js, holds ownership from UI to production infrastructure.

## Technical Skills

**AI & Integrations**: LLMs APIs, Vercel AI SDK, RAG, vector search, webhooks, REST APIs, background workers/job queues
**Languages**: TypeScript, JavaScript, Go, SQL | **Frameworks**: Node.js, Express.js, React.js, Next.js, Electron
**Databases**: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Drizzle ORM | **Frontend**: Tailwind CSS, shadcn/ui, Framer Motion
**DevOps/Infra**: Docker, GitHub Actions, Vercel, Firebase, Appwrite, Convex, Azure, AWS (basic)
**Testing & Tools**: Git, Jest, Vitest, React Testing Library

## Professional Experience

**Devspace** | *Software Engineer* *Mar 2026 – Aug 2026*

- Built full-stack React and TypeScript features on an **agentic AI pipeline**, including a **dynamic tool registry** where only relevant tools are fetched from the DB using **vector search**; contributed to CI/CD pipeline configuration and deployment automation.
- **Tech:** *React, React Native, TypeScript, Node.js, Convex, Vercel AI SDK, GitHub Actions, Codex*

**Osilion** | *Full-Stack Engineer* *Jul 2025 – Feb 2026*

- Served as one of the **primary engineers** on a client's internal employee productivity tool; contributed to the database design, **technical architecture**, and component design patterns of an intelligent recruitment platform, and mentored a junior developer through regular code reviews.
- **Tech:** *Next.js, TypeScript, Tailwind, shadcn/ui, Zustand, Appfarm (low-code), C#, Azure*

**Hone** | *Frontend Engineer* *Aug 2025 – Sep 2025*

- Implemented a **Smart Scaling system** that automatically adapts the desktop app to smaller screens, improved code splitting and lazy loading to cut initial load weight, and fixed a keyboard focus-visibility issue.
- **Tech:** *React, TypeScript, Electron, Tailwind, shadcn/ui, Claude Code*

**Hello World Communications** | *Full-Stack Engineer* *Sep 2024 – Jul 2025*

- **Led the development flow end to end** across multiple concurrent projects and made key architectural decisions; reduced client closing time by **roughly 50%** through better scoping, and mentored a junior developer across multiple client projects.
- **Tech:** *Next.js, TypeScript, Node.js, Go, PostgreSQL, Drizzle ORM, shadcn/ui, TanStack Query, Zustand, Firebase, Docker*

**Bilsida** | *Full-Stack Engineer* *Oct 2023 – Aug 2024*

- Built a car marketplace **from scratch across the full stack**, owning backend logic, **database design**, and image optimization; it went on to attract **two of Sweden's largest car dealers**. Also mentored a junior developer in building a chat system.
- **Tech:** *Next.js, TypeScript, Node.js, Tailwind, shadcn/ui, Appwrite, Docker*

**Fiverr** | *Freelance Full-Stack Developer* *Jul 2021 – Oct 2023*

- Delivered full-stack web applications and APIs for **20+ clients**, covering frontend, backend, and deployment.

## Projects

**Spud** Live | Source

- AI-powered Discord bot that detects **duplicate tasks**, flags **scope drift** and auto-closes tasks on PR merge; cut LLM token usage by an estimated **∼10x** with **vector search**, and built a **GitHub App** webhook integration along with Grafana monitoring.
- **Tech:** *TypeScript, Bun, discord.js, SQLite, GitHub Apps API, GitHub Actions, Gemini, Vercel AI SDK, Grafana, Docker*

**Narrative Guard** Live | Source

- Workspace-based brand governance SaaS platform with multi-tenant isolation, role-based access, and an end-to-end AI audit flow.
- **Tech:** *Next.js, TypeScript, Convex, Gemini, RAG, shadcn/ui, Vitest, React Testing Library*

**PGSketch** Source

- Linux desktop app that turns visual **Postgres** schema edits into reviewed **migration SQL**: a diagram editor with live change highlighting, risk warnings on destructive changes, a rolled-back dry-run validator, and drift detection when the database changes underneath you. Read-only by design, supports Postgres 14+, and ships as a `.deb`.
- **Tech:** *Electron, React, TypeScript, PostgreSQL, SQLite, Zod, Vitest, Playwright, Docker*

**DocTrac** Source

- A secure clinic admin portal to manage doctors and patients, with authentication, search and filtering, pagination, and a live analytics dashboard.
- **Tech:** *React, TypeScript, Express, MongoDB, Mongoose, Zod, TanStack Query, shadcn/ui, JWT*

**URL Health Checker** Source

- Built a **background job queue** in Go with separate API and worker services, concurrency-safe job claiming, **exponential backoff**, stale-job recovery, and integration tests against real PostgreSQL.
- **Tech:** *Go, PostgreSQL, Docker, pgx/v5, golang-migrate, slog*

**Result Lookup** Source

- Built a high-throughput exam result system **serving 2M results** at **∼32k RPS** peak, with **p99 under 105 ms** at 1,000 concurrent connections. Tested with *hey*
- **Tech:** *Go, PostgreSQL, Redis, Docker, pgx/v5, golang-migrate, slog, miniredis*

## Education

**BGC Trust University** | *B.Sc. in Computer Science and Engineering* *Chattogram, Bangladesh*